

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417097
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Yokono; Tomoya et al.

Accelerator control system, accelerator control method and accelerator control program

Abstract

An accelerator control system includes an accelerator control device and an accelerator, wherein the accelerator control device includes first processing circuitry configured to store control data including a location of data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator, and determine completion of the arithmetic processing by the accelerator when the control data which has been subjected to the arithmetic processing by the accelerator is stored in a storage, and the accelerator includes second processing circuitry configured to acquire the control data from the storage, perform arithmetic processing on the data which is an arithmetic processing target according to the location of the data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator included in the acquired control data, and store the control data in the storage when the arithmetic processing is completed.

Inventors:	Yokono; Tomoya (Musashino, JP), Yamabe; Yoshiro (Musashino, JP), Ishizaki; Teruaki (Musashino, JP)
Applicant:	NIPPON TELEGRAPH AND TELEPHONE CORPORATION (Tokyo, JP)
Family ID:	1000008818057
Assignee:	NIPPON TELEGRAPH AND TELEPHONE CORPORATION (Tokyo, JP)
Appl. No.:	18/287193
Filed (or PCT Filed):	April 22, 2021
PCT No.:	PCT/JP2021/016314
PCT Pub. No.:	WO2022/224410
PCT Pub. Date:	October 27, 2022

Prior Publication Data

Document Identifier

US 20240201986 A1

Publication Date

Jun. 20, 2024

Publication Classification

Int. Cl.: G06F9/30 (20180101); G06F9/28 (20060101)

U.S. Cl.:

CPC G06F9/28 (20130101); G06F9/3001 (20130101); G06F9/30043 (20130101); G06F2209/509 (20130101)

Field of Classification Search

CPC: G06F (9/28); G06F (9/3001); G06F (9/30043); G06F (2209/509)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11221848	12/2021	Maiyuran	N/A	G06F 17/16
11281609	12/2021	Kaneko	N/A	G06F 9/3836
2019/0042091	12/2018	Raghunath	N/A	G11C 29/028
2024/0202033	12/2023	Yokono	N/A	G06F 9/5038

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2019-149086	12/2018	JP	N/A
2020-537235	12/2019	JP	N/A
2019/073193	12/2018	WO	N/A

OTHER PUBLICATIONS

Yang et al., "When Poll is Better than Interrupt", FAST, vol. 12, Available Online at: <https://www.usenix.org/system/files/conference/fast12/yang.pdf>, 2012, pp. 1-7. cited by applicant

"CUDA Toolkit Documentation v11.2.2", NVIDIA, Available Online at: <https://docs.nvidia.com/cuda/>, Retrieved from the net on: Mar. 29, 2021, 1 page. cited by applicant

Gulati et al., "GPU Architecture and the CUDA Programming Model", Hardware acceleration of EDA algorithms, Available Online at: https://link.springer.com/content/pdf/10.1007%2F978-1-4419-0944-2_3.pdf, 2010, pp. 23-30. cited by applicant

Matthias Noack, et al., "Heterogeneous Active Messages for Offloading on the NEC SX-Aurora TSUBASA", 2019 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), May 20, 2019, 10 pages, XP033583415. cited by applicant

Lena Oden, et al., "Infiniband-Verbs on GPU: a case study of controlling an Infiniband network device from the GPU", 2014 IEEE 28th International Parallel & Distributed Processing Symposium Workshops, May 19, 2014, 8 pages, XP032696287. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is based on PCT filing PCT/JP2021/016314, filed Apr. 22, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to an accelerator control system, an accelerator control method and an accelerator control program.

BACKGROUND ART

(3) Conventionally, technology for increasing a processing speed by incorporating arithmetic accelerators into an arithmetic processing system is known. Since it is difficult to improve the operating frequency of a processor and to highly integrate arithmetic units due to the limitation of semiconductor microfabrication technology, processing is offloaded to an accelerator in a specialized manner for specific processing by this technology, and high-performance calculation is realized (refer to NPL 1 to 3).

(4) Specifically, a CPU performs control for driving an accelerator via an API for the accelerator and a device driver, and the accelerator reads processing data copied to a kernel memory space, and performs arithmetic processing which takes time to be executed by the CPU. Further, the accelerator notifies the CPU of completion of processing by hardware interruption after execution of the arithmetic processing.

CITATION LIST

Non Patent Literature

(5) [NPL 1] J. Yang, D. B. Minturn, F. Hady, "When Poll is Better than Interrupt," FAST, vol. 12, pp. 3-3, [online], 2012, [Retrieved on Mar. 29, 2021], Internet <URL: usenix.org/system/files/conference/fast12/yang.pdf> [NPL 2] "CUDA Toolkit Documentation v11.2.2, [online], NVIDIA, [Retrieved on Mar. 29, 2021], Internet <URL: <https://docs.nvidia.com/cuda/>> [NPL 3] K. Gulati, S. P. Khatri, "GPU Architecture and the CUDA Programming Model," Hardware acceleration of EDA algorithms, pp. 23-30, [online], Springer, [Retrieved on Mar. 29, 2021], Internet <URL: https://ink.springer.com/content/pdf/10.1007%2F978-1-4419-0944-2_3.pdf>

SUMMARY OF INVENTION

Technical Problem

(6) However, in the conventional technology, it is sometimes difficult to increase the speed of real-time processing using accelerators. For example, since accelerators are controlled by the CPU, each time processing such as writing of data to an accelerator, execution of arithmetic operation, or reading of data is performed, interrupt notification of exclusive control and processing completion is generated. In addition, a memory copy occurs in a user space and a kernel space for reading and writing data. Such processing becomes an overhead and increases a delay time. When there are a plurality of accelerators, an overhead occurs in the same manner, and scalability is reduced. Therefore, it is difficult to use accelerators in the area of real-time processing.

(7) The present invention has been made in view of the foregoing and an object thereof is to increase the speed of real-time processing using accelerators.

Solution to Problem

(8) In order to solve the aforementioned problems and achieve the object, an accelerator control system includes an accelerator control device and an accelerator, wherein the accelerator control

device includes: first processing circuitry configured to: store control data including a location of data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator; and determine completion of the arithmetic processing by the accelerator when the control data which has been subjected to the arithmetic processing by the accelerator is stored in a storage, and the accelerator includes: second processing circuitry configured to: acquire the control data from the storage; perform arithmetic processing on the data which is an arithmetic processing target according to the location of the data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator included in the acquired control data; and store the control data in the storage when the arithmetic processing is completed.

Advantageous Effects of Invention

(9) According to the present invention, it is possible to increase the speed of real-time processing using accelerators.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating an overall configuration of an accelerator control system of the present embodiment.
- (2) FIG. 2 is a diagram for describing the accelerator control system of the present embodiment.
- (3) FIG. 3 is a diagram illustrating a data configuration of metadata.
- (4) FIG. 4 is a diagram illustrating a structure of a metadata buffer.
- (5) FIG. 5 is a diagram for describing a data buffer.
- (6) FIG. 6 is a flowchart showing an accelerator control processing procedure in an accelerator control device.
- (7) FIG. 7 is a flowchart showing an accelerator control processing procedure in an accelerator.
- (8) FIG. 8 is a sequence diagram showing an accelerator control processing procedure.
- (9) FIG. 9 is a diagram for describing an example.
- (10) FIG. 10 is a diagram showing an example of a computer that executes an accelerator control program.

DESCRIPTION OF EMBODIMENTS

(11) An embodiment of the present invention will be described in detail below with reference to the drawings. Note that the present invention is not limited to the embodiment. Furthermore, the same constituent elements are denoted by the same reference numerals in the description of the drawings.

(12) [Configuration of Accelerator Control System]

(13) FIG. 1 is a schematic diagram illustrating an overall configuration of an accelerator control system of the present embodiment. As shown in FIG. 1, the accelerator control system 1 includes an accelerator control device 10 and an accelerator 20. The accelerator control device 10 and the accelerator 20 can communicate data with each other via a communication line such as bus wiring, for example.

(14) The accelerator control device 10 is an information processing device realized using a general-purpose processor such as a central processing unit (CPU). In the accelerator control device 10, an application which is processing defined by a user is executed in a user space. In addition, operation and management of a physical device such as the accelerator 20 and abstraction processing of the physical device in user space processing are performed in a kernel space.

(15) A user memory space and a kernel memory space are realized by a semiconductor memory element such as a random access memory (RAM) or a flash memory, or a storage device such as a hard disk or an optical disc and serve as a storage unit. A processing program for operating the accelerator control device 10, data used during execution of the processing program, and the like

are stored in advance or stored temporarily each time processing is performed in the storage unit.

(16) A control unit realized using a general-purpose processor serves as each functional unit such as a memory management function **12** and a metadata control function **13** of a device driver, or an API **14** which will be described later by executing the processing program stored in the memory, as illustrated in FIG. **1**. Note that the control unit may include another functional unit that is not shown.

(17) In the accelerator control device **10**, data that is an arithmetic processing target (hereinafter also referred to as processing data) is disposed in a data buffer **11a** of the user memory space on the user space. Further, control data (hereinafter also referred to as metadata) including the location of data that is an arithmetic processing target and information specifying the content of arithmetic processing of an accelerator is disposed in a metadata buffer **11b** of the kernel memory space in the kernel space.

(18) The accelerator **20** is an arithmetic accelerator realized by a graphics processing unit (GPU), a field programmable gate array (FPGA), a data processing unit (DPU) or the like, and as illustrated in FIG. **1**, serves as functional units such as a metadata access mechanism **21**, a function control block **22**, a data transfer function **23**, and an arithmetic function **24**.

(19) The accelerator **20** reads processing data and metadata to realize offload of arithmetic processing. Specifically, the accelerator **20** monitors whether or not metadata is disposed in the metadata buffer **11b** according to autonomous polling. When the metadata is disposed therein, the accelerator **20** reads the disposed metadata, analyzes it, and drives the accelerator **20**. In addition, the accelerator **20** stores the result of arithmetic processing performed on the processing data in the data buffer **11a**.

(20) The accelerator control device **10** can proceed to the next processing asynchronously with the accelerator **20**. The accelerator **20** disposes the metadata in the metadata buffer **11b** when the arithmetic processing is completed. Accordingly, the accelerator control device **10** considers that the arithmetic processing has been completed. Further, the accelerator control device **10** acquires the result of arithmetic processing performed on the processing data from the data buffer **11a** and proceeds to the next processing.

(21) Specifically, the accelerator control device **10** determines the operation of the accelerator **20** by arranging metadata including the location of processing data and the content of arithmetic processing in the metadata buffer **11b** in the kernel memory space through the metadata control function **13** mounted in the device driver.

(22) The accelerator control device **10** creates a data buffer **11a** which can be referred to by the accelerator control device **10** and the accelerator **20** in the user memory space via the memory management function **12**, and disposes the processing data.

(23) The accelerator **20** has a metadata access mechanism **21**. The metadata access mechanism **21** has a function of polling the metadata buffer **11b** in the kernel memory space, a function of reading metadata, and a function of writing metadata into the metadata buffer, and can autonomously read metadata without depending on an explicit control signal from the accelerator control device **10**.

(24) The accelerator **20** is notified of information such as an address of the metadata buffer **11b** by an accelerator initialization function at the time of loading the device driver.

(25) The accelerator **20** reads metadata from a metadata buffer **11b**, and a metadata analysis function **22a** analyzes the data and operates according to the description. The accelerator **20** drives a data reading function **23a** in the data transfer function **23** on the basis of information on the address and length of the processing data included in the metadata, reads the processing data from the data buffer **11a**, and uses it for an arithmetic operation in the arithmetic function **24**. In the arithmetic function **24**, arithmetic processing is executed on the basis of arithmetic information written in the metadata. The metadata analysis function **22a** transfers the analysis result to the arithmetic function **24**, and sets the content of arithmetic processing by the arithmetic function **24**.

(26) After arithmetic processing is completed, the result of arithmetic processing is transferred from

the arithmetic function **24** to a data writing function **23b** and written in a data buffer designated by a data transfer control function **22b**. Thereafter, a notification indicating that data transfer is completed is transmitted from the data transfer function **23** to a completion notification function **22c**, the metadata is written in the metadata buffer **11b** by a writing function of the metadata access mechanism **21**, and the operation of the accelerator **20** is completed.

(27) In this manner, in the accelerator control system **1**, metadata for controlling data transfer and arithmetic processing is disposed in a main memory, and the accelerator **20** autonomously performs polling and reading. Accordingly, an overhead of accelerator control such as interruption is reduced, and real-time processing can be performed.

(28) In addition, since the accelerator **20** can directly transfer processing data to the user space of the accelerator control device **10** and the accelerator control device **10** and the accelerator **20** can be driven asynchronously, delay and CPU use rate are reduced.

(29) Further, since the accelerator **20** can operate by accessing the main memory of the accelerator control device **10**, extension to a plurality of accelerators **20** can be achieved and thus scalability is greatly improved.

(30) [Accelerator Control Device] FIG. **2** is a diagram for describing the accelerator control system of the present embodiment. As illustrated in FIG. **2**, in the accelerator control device **10**, the API **14** is a group of functions that can be called through an application and provides functions necessary for the operation of the accelerator **20**. For example, the API serves as a data buffer securing function, a data buffer releasing function, a metadata creation function, a metadata transmission function, and an arithmetic operation completion sensing function.

(31) The data buffer securing function generates the data buffer **11a** which is a storage area for transmitting/receiving data to/from the accelerator **20**. The data buffer securing function calls the memory management function **12** and acquires area information of the data buffer **11a** from the memory management function **12**.

(32) The data buffer release function deletes the data buffer **11a**. The data buffer release function calls the memory management function **12**, notifies of the area information of the data buffer **11a** to be released, and releases the area.

(33) The metadata creation function creates metadata that is control data to which control information of the accelerator **20** has been added. The metadata transmission function transmits the metadata created by the metadata creation function to the metadata control function **13**. The arithmetic operation completion sensing function calls the metadata control function **13** and checks whether or not processing of an arbitrary accelerator **20** is completed according to a return value of the metadata control function **13**.

(34) The device driver is present in the kernel space and provides a function group such as the memory management function **12** and the metadata control function **13** for handling a specific device.

(35) The memory management function **12** is called from the API **14** in the user space, and secures, releases and manages the area of the data buffer **11a**.

(36) The metadata buffer **11b** is a storage area for transmitting/receiving metadata to/from the accelerator **20** and is generated depending on the number of accelerators **20** by an accelerator initialization function. For example, the metadata buffer **11b** stores metadata including the location of data that is an arithmetic processing target and information specifying the content of arithmetic processing of the accelerator. Further, the metadata buffer **11b** stores different metadata for each of the plurality of accelerators **20**.

(37) The metadata buffer **11b** is composed of a request queue (RQ) buffer for transmitting metadata to the accelerator **20** and a completion queue (CQ) buffer for receiving metadata from the accelerator **20**.

(38) The metadata control function **13** is called from the API **14** in the user space and performs reading, writing and management with respect to the RQ buffer/CQ buffer of the metadata buffer

11b. The metadata control function **13** has a polling function **13a** and a writing function **13b**.

(39) The polling function **13a** polls the CQ buffer and checks completion of an arithmetic operation. That is, the polling function **13a** serves as a determination unit to monitor whether or not metadata that has been subjected to arithmetic processing by the accelerator **20** is stored in the metadata buffer **11b**. Then, the polling function **13a** determines completion of the arithmetic processing by the accelerator **20** when the metadata that has been subjected to the arithmetic processing by the accelerator **20** has been stored in the metadata buffer **11b**. Further, the writing function **13b** writes data into the RQ buffer.

(40) [Accelerator] In the accelerator **20**, the metadata access mechanism **21** includes, as functions for accessing the metadata buffer **11b** in the kernel space, a polling function **21a**, a reading function **21b**, and a writing function **21c**.

(41) The polling function **21a** polls the RQ buffer of the metadata buffer **11b** to check whether or not metadata is present in the RQ buffer. That is, the polling function **21a** serves as a monitoring unit to monitor whether or not metadata is present in the metadata buffer **11b**.

(42) The reading function **21b** serves as an acquisition unit to acquire metadata from the metadata buffer **11b**. That is, the reading function **21b** reads metadata and transfers it to the function control block **22** when the polling function **21a** confirms that the metadata is present in the RQ buffer.

(43) The writing function **21c** writes metadata in the CQ buffer of the metadata buffer **11b**. That is, the writing function **21c** serves as a storage unit, and when arithmetic processing is completed, stores metadata in the metadata buffer **11b**.

(44) The function control block **22** controls sharing, driving and pause timing of necessary data for each functional unit of the accelerator **20**. The function control block **22** includes the metadata analysis function **22a**, the data transfer control function **22b**, and the completion notification function **22c**.

(45) The metadata analysis function **22a** extracts information on an operation, and the address and length of the data buffer **11a** from metadata transmitted from the metadata access mechanism **21**. The data transfer control function **22b** provides a function for controlling reading and writing of data in the data transfer function **23**. The completion notification function **22c** transfers metadata to be written in the CQ buffer to the metadata access mechanism **21**.

(46) The arithmetic function **24** serves as an arithmetic unit to perform arithmetic processing on processing data according to information specifying the location of the processing data and the operation of the accelerator **20** included in the acquired metadata.

(47) Specifically, the arithmetic function **24** includes an input control function, an arithmetic circuit, and an output control function. The arithmetic circuit has a plurality of independent arithmetic circuits and can define an arithmetic operation for each thereof. The input control function inputs data transmitted from the data reading function **23a** of the data transfer function **23** to an appropriate arithmetic circuit on the basis of information on metadata transmitted from the function control block **22**.

(48) The output control function transfers the result of arithmetic processing performed on the processing data output from the arithmetic circuit to the data writing function **23b** of the data transfer function **23** at appropriate timing.

(49) The data writing function **23b** of the data transfer function **23** serves as a storage unit to store the result of arithmetic processing performed on the processing data in the data buffer **11a**.

(50) Next, FIG. **3** is a diagram illustrating a data configuration of metadata. As shown in FIG. **3**, the metadata is composed of a total of 256-bit data including a 32-bit task ID, a 32-bit operation, a 32-bit read data length, a 64-bit read data address, a 32-bit write data length, and a 64-bit write data address.

(51) FIG. **4** is a diagram illustrating the structure of the metadata buffer. As shown in FIG. **4**, the metadata shown in FIG. **3** is stored in the RQ buffer/CQ buffer. Head information of the RQ buffer shown in FIG. **4** is controlled by the metadata control function **13**. The metadata control function

13 forms the metadata transmitted from the metadata transmission function of the API **14** into the configuration shown in FIG. **3**, stores the metadata in the RQ buffer, and then shifts the head information to the next position.

(52) Similarly, with respect to the CQ buffer, the metadata control function **13** also reads metadata from the CQ buffer using the head information and shifts the head to the next position when called by the arithmetic operation completion sensing function of the API **14**.

(53) FIG. **5** is a diagram for describing the data buffer. The memory management function **12** manages the address and status of the data buffer **11a** through a table illustrated in FIG. **5**. As shown in FIG. **5**, the physical address of the data buffer **11a** and the status of the address are managed through the table. At the time of setting metadata for the accelerator **20**, a physical address of a desired data buffer **11a** is acquired and a read data address and a write data address are set by referring to a key. Accordingly, differences between virtual addresses in the kernel space and the user space can be absorbed and managed.

(54) [Accelerator Control Processing] Next, accelerator control processing performed by the accelerator control system **1** according to the present embodiment will be described with reference to FIG. **6** to FIG. **8**. FIG. **6** is a flowchart showing an accelerator control processing procedure in the accelerator control device **10**. The flowchart of FIG. **6** is started at a timing at which, for example, an application instructs starting.

(55) First, the accelerator control device **10** secures the data buffer **11a** (step S1). The accelerator control device **10** stores (disposes) processing data to be transferred to the accelerator **20** in the data buffer **11a** (step S2).

(56) The accelerator control device **10** sets information on metadata (step S3) and submit the information to an RQ buffer of the metadata buffer **11b** (step S4). Then, the accelerator control device **10** executes processing independent of the arithmetic result of the accelerator **20** (step S5).

(57) When the accelerator control device **10** uses the arithmetic result of the accelerator **20**, the accelerator control device **10** accesses a CQ buffer of the metadata buffer **11b** and checks whether or not the accelerator **20** transmits metadata (step S6). If there is no metadata in the CQ buffer (No in step S6), the accelerator control device **10** returns processing to step S5. On the other hand, if metadata is present in the CQ buffer (Yes in step S6), the accelerator control device **10** determines notification of completion of the arithmetic processing, acquires the arithmetic result of the processing data from the data buffer **11a**, and continues processing (step S7).

(58) If the accelerator **20** is reused (No in step S8), the accelerator control device **10** returns processing to step S2. In this case, the data buffer **11a** can be reused. On the other hand, if it is not necessary to reuse the accelerator **20** (Yes in step S8), the accelerator control device **10** releases the area of the data buffer **11a** when the application ends (step S9). Accordingly, a series of processing ends.

(59) FIG. **7** is a flowchart showing an accelerator control processing procedure in the accelerator **20**. The flowchart of FIG. **7** is started at predetermined intervals, for example. The accelerator **20** monitors the RQ buffer of the metadata buffer **11b** by polling (step S11).

(60) If metadata is not set in the RQ buffer (No in step S12), the accelerator **20** returns processing to step S11. On the other hand, if metadata is set in the RQ buffer (Yes in step S12), the accelerator **20** reads the metadata and analyzes it (step S13).

(61) The accelerator **20** acquires processing data of the data buffer **11a** on the basis of the analysis result (step S14), and performs arithmetic processing (step S15). Next, the accelerator **20** writes the processing data that has been subjected to arithmetic processing in the data buffer **11a** (step S16), transmits the metadata to the CQ buffer (step S17), and ends processing. Thereafter, the accelerator **20** returns processing to step S11.

(62) Next, FIG. **8** is a sequence diagram showing an accelerator control processing procedure. In the example shown in FIG. **8**, the accelerator **20** is implemented by an FPGA. As shown in FIG. **8**, first, the accelerator control device **10** disposes processing data for arithmetic processing in a

shared memory space (data buffer **11a**) (step **S21**). In addition, the accelerator control device **10** disposes metadata for the accelerator **20** in an RQ buffer of the metadata buffer **11b** in the kernel memory space via the device driver (step **S22**).

(63) The accelerator **20** monitors whether or not metadata is disposed by polling the RQ buffer (step **S23**). The accelerator **20** reads the metadata when the metadata is disposed, analyzes processing content written in the metadata and the position of processing data, and reads the processing data (step **S24**).

(64) The accelerator **20** writes the processing data that has been subjected to arithmetic processing in the data buffer **11a** (step **S25**), and writes the metadata in a CO buffer of the metadata buffer **11b** (step **S26**).

(65) The accelerator control device **10** accesses the CQ buffer to check whether or not metadata is disposed, and senses completion of arithmetic processing of the accelerator **20** when the metadata is disposed (step **S27**). In this case, the accelerator control device **10** acquires data of the arithmetic processing result from the data buffer **11a** (step **S28**).

(66) [Effects] As described above, in the accelerator control system **1** of the present embodiment, the metadata buffer **11b** stores metadata including the location of processing data which is an arithmetic processing target and information specifying the content of arithmetic processing of the accelerator **20** in the accelerator control device **10**. The polling function **13a** determines completion of the arithmetic processing performed by the accelerator **20** when metadata that has been subjected to the arithmetic processing performed by the accelerator **20** is stored in the metadata buffer **11b**. In the accelerator **20**, the reading function **21b** acquires the metadata from the metadata buffer **11b**. The arithmetic function **24** performs arithmetic processing on processing data that is an arithmetic processing target according to the location of the processing data that is an arithmetic processing target and information specifying the operation of the accelerator **20** included in the acquired metadata. When the arithmetic processing is completed, the writing function **21c** stores the metadata in the metadata buffer **11b**.

(67) Accordingly, the accelerator control system **1** can perform real-time processing at a high speed by reducing an overhead such as exclusive control, interrupt notification of completion of processing, and memory copy in processing using the accelerator **20**.

(68) Further, the polling function **21a** monitors whether or not metadata is present in the metadata buffer **11b**. Accordingly, the accelerator **20** can autonomously perform processing.

(69) Further, the polling function **13a** monitors whether or not metadata that has been subjected to arithmetic processing is present in the metadata buffer **11b**. Accordingly, the accelerator control device **10** can perform processing asynchronously with the accelerator **20**.

(70) The data writing function **23b** stores the result of arithmetic processing performed on processing data that is an arithmetic processing target in the data buffer **11a**. Accordingly, the accelerator control device **10** can perform processing asynchronously with the accelerator **20** except cases where the arithmetic processing result of the accelerator **20** is used, and thus the speed of processing using the accelerator **20** can be increased.

(71) Further, the metadata buffer **11b** stores metadata different for each of the plurality of accelerators **20**. Accordingly, scalability according to the plurality of accelerators **20** is improved.

(72) [Examples] FIG. **9** is a diagram for describing an example. FIG. **9** illustrates measured values of time until processing data returns through loopback from the accelerator control device **10** via the accelerator **20** in processing (proposed method) by the accelerator control system **1** of the above-described embodiment and processing (conventional method) by the conventional technology. In this case, as bus technology, PCIe Gen 3×8 is used in the conventional method, and PCIe Gen 3×16 is used in the proposed method.

(73) FIG. **9** shows that, for example, latency is 20 usec in the proposed method whereas latency is 160 usec or more in the conventional method with respect to 128 Kbyte data. In this manner, it has been confirmed that the proposed method can more increase a processing speed than the

conventional method.

(74) [Program] It is also possible to create a program in which processing executed by the accelerator control device **10** according to the above-described embodiment is written in a language executable by a computer. As one embodiment, the accelerator control device **10** can be implemented by installing an accelerator control program for executing the aforementioned accelerator control processing as package software or online software in a desired computer. For example, by causing an information processing device to execute the accelerator control program, the information processing device can be caused to serve as the accelerator control device **10**. The information processing device mentioned here includes a desktop type or laptop type personal computer. In addition, the information processing device includes a mobile communication terminal such as a smartphone, a mobile phone or a personal handyphone system (PHS), a slate terminal such as a personal digital assistant (PDA), or the like. Furthermore, the functions of the accelerator control device **10** may be implemented in a cloud server.

(75) FIG. **10** is a diagram showing an example of a computer that executes the accelerator control program. A computer **1000** has a memory **1010**, a CPU **1020**, a hard disk drive interface **1030**, a disc drive interface **1040**, a serial port interface **1050**, a video adapter **1060**, and a network interface **1070**, for example. These units are connected through a bus **1080**.

(76) The memory **1010** includes a read only memory (ROM) **1011**, and a RAM **1012**. The ROM **1011** stores, for example, a boot program such as a basic input output system (BIOS). The hard disk drive interface **1030** is connected to a hard disk drive **1031**. The disc drive interface **1040** is connected to the disc drive **1041**. A removable storage medium such as a magnetic disk or an optical disc is inserted into the disc drive **1041**. For example, a mouse **1051** and a keyboard **1052** are connected to the serial port interface **1050**. For example, a display **1061** is connected to the video adapter **1060**.

(77) Here, the hard disk drive **1031** stores, for example, an OS **1091**, an application program **1092**, a program module **1093**, and program data **1094**. Each of the pieces of information described in the above embodiment is stored in, for example, the hard disk drive **1031** or the memory **1010**.

(78) Further, the accelerator control program is stored in the hard disk drive **1031** as the program module **1093** in which commands executed by the computer **1000** are written, for example. Specifically, the program module **1093** in which each processing executed by the accelerator control device **10** described in the foregoing embodiment is written is stored in the hard disk drive **1031**.

(79) The data used for information processing by the accelerator control program is stored in, for example, the hard disk drive **1031** as the program data **1094**. Thereafter, the CPU **1020** reads the program module **1093** and the program data **1094** stored in the hard disk drive **1031** into the RAM **1012** when necessary, and executes each of the procedures described above.

(80) Note that the program module **1093** and program data **1094** related to the accelerator control program are not limited to being stored in the hard disk drive **1031**, and may also be stored in, for example, a removable storage medium and read out by the CPU **1020** via the disc drive **1041**, etc.

(81) Alternatively, the program module **1093** and the program data **1094** related to the accelerator control program may be stored in another computer connected via a network such as a LAN or a wide area network (WAN), and may be read by the CPU **1020** via the network interface **1070**.

(82) Although the embodiments to which the invention made by the present inventor is applied have been described above, the present invention is not limited by the descriptions and drawings forming a part of the disclosure of the present invention according to the embodiments. That is to say, other embodiments, examples, operation techniques, and the like made by those skilled in the art on the basis of the embodiment are all included in the scope of the present invention.

REFERENCE SIGNS LIST

(83) **10** Accelerator control device **11a** Data Buffer (storage unit) **11b** Metadata buffer (storage unit) **12** Memory management function **13** Metadata control function **13a** Polling function

(determination unit) **13b** Writing function **14** API **20** Accelerator **21** Metadata access mechanism **21a** Polling function (acquisition unit) **21b** Reading function **21c** Writing function (storage unit) **22** Function control block **22a** Metadata analysis function **22b** Data transfer control function **22c** Completion notification function **23** Data transfer function **23a** Data reading function **23b** Data writing function (storage unit) **24** Arithmetic function (arithmetic unit)

Claims

1. An accelerator control system comprising an accelerator control device and an accelerator, wherein the accelerator control device includes: first processing circuitry configured to: store control data including a location of data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator; and determine completion of the arithmetic processing by the accelerator when the control data which has been subjected to the arithmetic processing by the accelerator is stored in a storage, and the accelerator includes: second processing circuitry configured to: acquire the control data from the storage; perform arithmetic processing on the data which is an arithmetic processing target according to the location of the data which is an arithmetic processing target and information specifying an operation of the accelerator included in the acquired control data; and store the control data in the storage when the arithmetic processing performed by the second processing circuitry is completed.
2. The accelerator control system according to claim 1, wherein the second processing circuitry is further configured to monitor whether or not the control data is present in the storage.
3. The accelerator control system according to claim 1, wherein the first processing circuitry is further configured to monitor whether or not the control data which has been subjected to the arithmetic processing by the second processing circuitry is present in the storage.
4. The accelerator control system according to claim 1, wherein the second processing circuitry is further configured to store, in the storage, a result of the arithmetic processing performed on the data by the second processing circuitry which is an arithmetic processing target.
5. The accelerator control system according to claim 1, wherein the first processing circuitry is further configured to store control data different for each of a plurality of accelerators.
6. An accelerator control method, comprising: acquiring control data, the control data including a location of data which is an arithmetic processing target and information specifying content of arithmetic processing of an accelerator; performing arithmetic processing on the data which is an arithmetic processing target according to the location of the data which is an arithmetic processing target and information specifying an operation of the accelerator included in the acquired control data; storing the control data when the arithmetic processing is completed; and determining completion of the arithmetic processing when the control data which has been subjected to the arithmetic processing is stored.
7. A non-transitory computer-readable recording medium storing therein an accelerator control program that causes a computer to execute a process comprising: acquiring control data including a location of data which is an arithmetic processing target and information specifying content of arithmetic processing of the accelerator; performing arithmetic processing on the data which is an arithmetic processing target according to the location of the data which is an arithmetic processing target and information specifying an operation of the accelerator included in the acquired control data; storing the control data when the arithmetic processing is completed; and determining completion of the arithmetic processing when the control data which has been subjected to the arithmetic processing is stored.
8. The accelerator control method according to claim 6, further comprising: monitoring whether or not the control data has been stored.
9. The accelerator control method according to claim 6, further comprising: monitoring whether or not the control data which has been subjected to the arithmetic processing has been stored.

10. The accelerator control method according to claim 6, further comprising: storing a result of the arithmetic processing performed on the data which is an arithmetic processing target.
 11. The accelerator control method according to claim 6, further comprising: storing control data which is different for each of a plurality of accelerators.
 12. The non-transitory computer-readable recording medium according to claim 7, wherein the processing further comprises: monitoring whether or not the control data has been stored.
 13. The non-transitory computer-readable recording medium according to claim 7, wherein the processing further comprises: monitoring whether or not the control data which has been subjected to the arithmetic processing has been stored.
 14. The non-transitory computer-readable recording medium according to claim 7, wherein the processing further comprises: storing a result of the arithmetic processing performed on the data which is an arithmetic processing target.
 15. The non-transitory computer-readable recording medium according to claim 7, wherein the processing further comprises: storing control data which is different for each of a plurality of accelerators.
-